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Section-A

Q.1) (B) Calcium and Magnesium

Q.2) (A) $Mg : \ddot{\underset{\cdot\cdot}{O}}: \rightarrow Mg^{2+}[:\ddot{\underset{\cdot\cdot}{O}}:^{2-}]$

Q.3) (C) It has weak electrostatic forces of attraction between its oppositely charged ions

Q.4) (A) Salt and water is formed

Q.5) (B) 5

Q.6) (B) Al_2O_3 and MgO

Q.7) (D) 1 : 8.

Q.8) (D) Cytoplasm and Oxygen deficient muscle cells

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Q.9) (C) 100% round and yellow

Q.10) (C) (i) and (iii)

Q.11) (D) Auxins

Q.12) (C) Starch into simple sugars

Q.13) (D) 99 %

Q.14) (D) (ii) and (iv)

Q.15) (A) scattering of light

Q.16) (A) (i) and (ii)

Q.17) (B) Both Assertion(A) and Reason(R) are true, but Reason (R) is not the correct explaination of (A).

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Q.18) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explaination of Assertion (A).

Q.19) (C) Assertion (A) is true, but Reason (R) is false.

Q.20) (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explaination of Assertion (A)

Section-B

Q.21) Given -

$u = -10\,cm$ (object distance)

$f = -15\,cm$ (since it is a concave mirror)

To find → v (image distance)

$\Rightarrow$ Mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{v} - \frac{1}{10} = \frac{-1}{15}$$

$$\Rightarrow \frac{1}{v} = \frac{-1}{15} + \frac{1}{10}$$

$$\Rightarrow \frac{1}{v} = \frac{-10+15}{10\times15}$$

$$\Rightarrow \frac{1}{v} = \frac{5}{\frac{10}{2}\times15}$$

$$\boxed{v = +30\text{cm}}$$

Position of image formed by mirror = +30cm on the other side of mirror.

Q.22)

(a) Lamp A →

Power = 50 ω

Volt = 220V

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We know,

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P}$$

$$R_A = \frac{220 \times 220}{50} = \frac{44}{} \times 22$$

$$= 44 \times 22$$

$$= 968\,\Omega$$ ✓

Lamp B →

Power = 25W

Volt = 220V

$$P = \frac{V^2}{R}$$

$$R_B = \frac{220 \times 220}{25} = \frac{44}{} \times \frac{44}{}$$

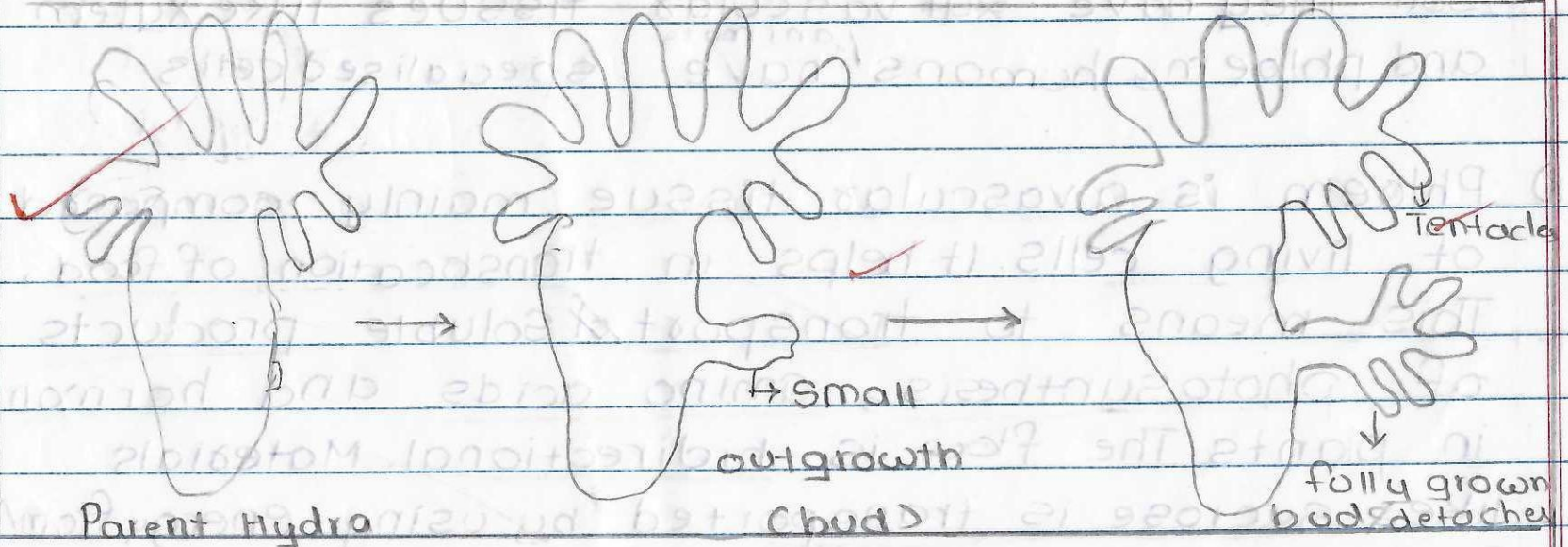
$$= 1936\,\Omega$$ ✓

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$$\therefore \frac{R_A}{R_B} = \frac{968}{1936}$$

$$R_A : R_B = \boxed{1 : 2}$$

(Q.23) Hydra reproduces by the process of budding. With help of regenerative cells, there is outgrowth called bud which develops.



Diagᵐ - Budding in Hydra

Space for writing Question Number

Q.24)

(b)(i) The transport system in plants is relatively slower than in animals because plants are mainly composed of dead cells. Plants are stationary and don't require energy to perform activities when compared to animals. Moreover, simple processes like diffusion and osmosis help in transport of materials from soil. They have vascular tissues like xylem and phloem, animals have specialised cells

(ii) Phloem is a vascular tissue mainly composed of living cells. It helps in translocation of food. This means to transport soluble products of photosynthesis, amino acids and hormones in plants. The flow is bidirectional. Materials like sucrose is transported by using energy from ATP for eg. During spring, food stored in stem

and roots is transported to buds.

Q.25) For a chemical change to occur there has to be-
i) evolution of gas
ii) change in temperature
iii) change in odour
iv) formation of precipitate

Here,

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2\uparrow$$

Here 2 observations are seen-
i) Evolution of gas (H_2)
ii) Change in temperature since this is an exothermic reaction.

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Q.26)

(a) $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$

(b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

Section-C

Q.27)

(a) When 1 Joule of work is done to move 1 Coloumb of charge from one point to another in an electric field, it is said to be 1 Volt

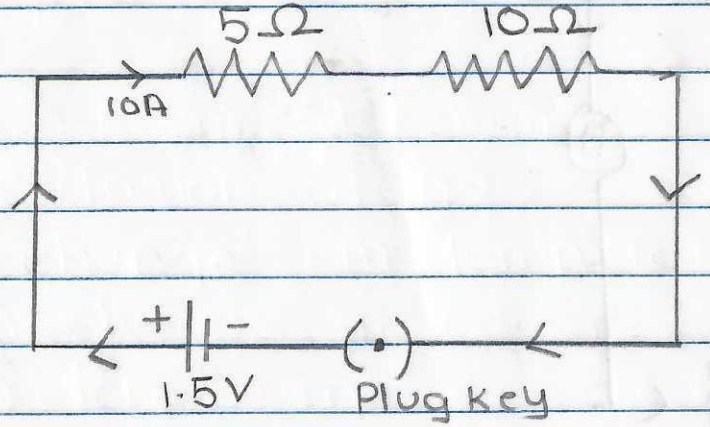
$$V = \frac{W}{Q}$$

$$1V = \frac{1J}{1C}$$

$$1V = 1JC^{-1}$$

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Q.27)
(b)



Let $R_1 = 5\Omega$

$R_2 = 10\Omega$

To find → I (current)

Total resistance (R_s) $= R_1 + R_2$

$= 5 + 10$

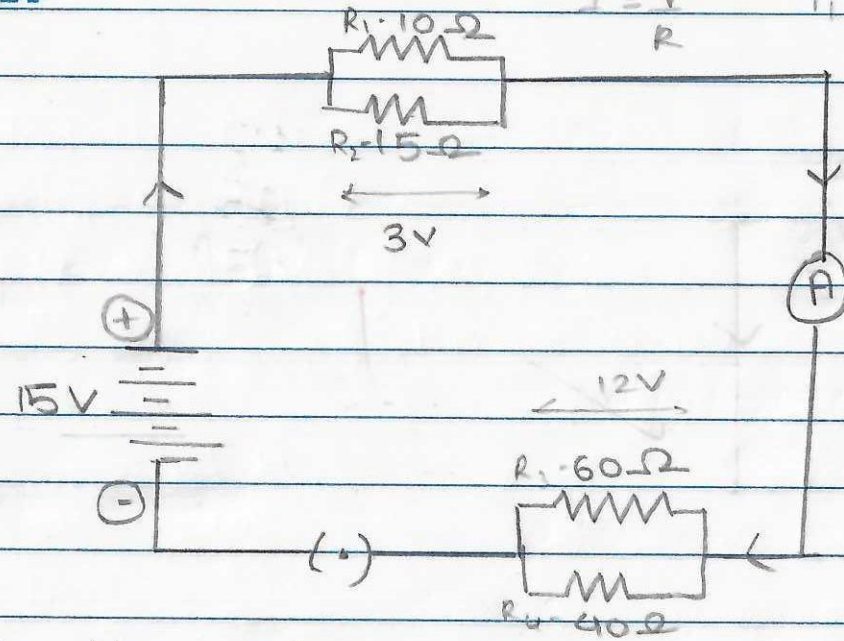
$= 15\Omega$

$I = \frac{V}{R}$ (Ohm's law)

$= \frac{1.5}{15} = \frac{1}{10}A$ **0.1A**

Space for writing Question Number

(Q.28)



In the given circuit,

Let $R_1 = 10\,\Omega$

$R_2 = 15\,\Omega$

$R_3 = 60\,\Omega$

$R_4 = 40\,\Omega$

(a) Since R_1 and R_2 are in parallel,

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_5}$$

$$\Rightarrow \frac{1}{10} + \frac{1}{15} = \frac{1}{R_5}$$

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$\Rightarrow \frac{15+10}{150} = \frac{1}{R_5}$

$\Rightarrow \frac{25}{150} = \frac{1}{R_5}$

30

$R_5 = 6\,\Omega$ ✓

Now R_3 and R_4 are in parallel,

$\frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{R_6}$

$\Rightarrow \frac{1}{60} + \frac{1}{40} = \frac{1}{R_6}$

$\Rightarrow \frac{100}{60 \times 40} = \frac{1}{R_6}$

$\Rightarrow \therefore R_6 = 24\,\Omega$ ✓

Now total $R_S = R_5 + R_6$

$= \boxed{30\,\Omega}$ → Total resistance ✓

(b) Total current →

$R = 30\Omega$

$V = 15V$

$I = \frac{V}{R}$ (Ohm's law)

$= \frac{1}{2} A$

(c) Total Resistance(R_5) of parallel combination of 10Ω and $15\Omega = 6\Omega$

$I = \frac{1}{2} A$

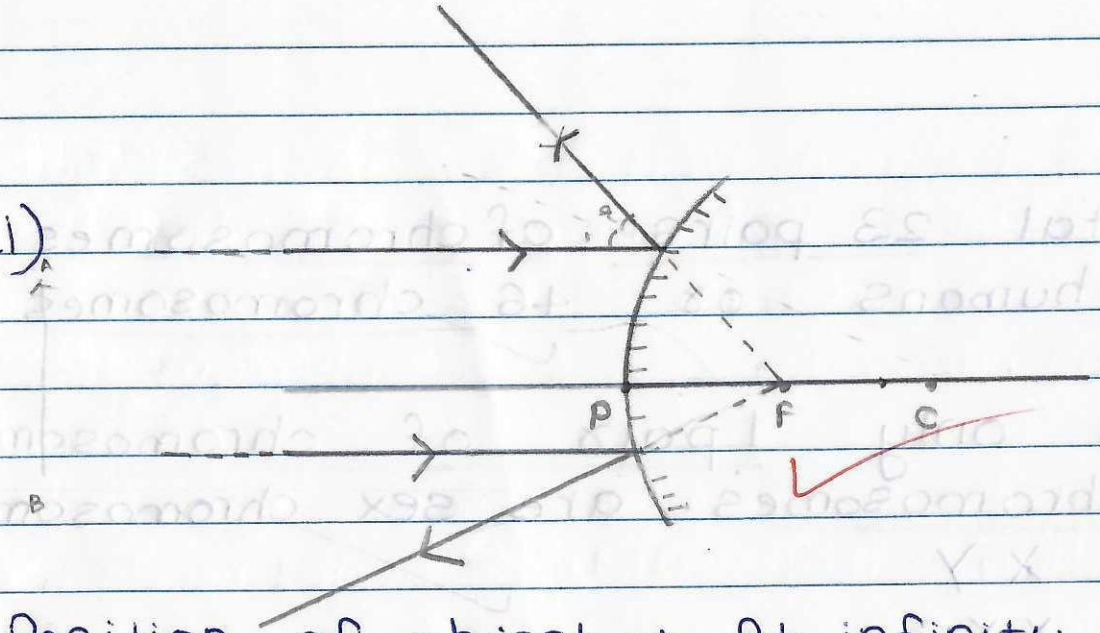
$\therefore V = IR$ (Ohm's law)

$= 3V$

Voltage across them is $3V$

Q 29)

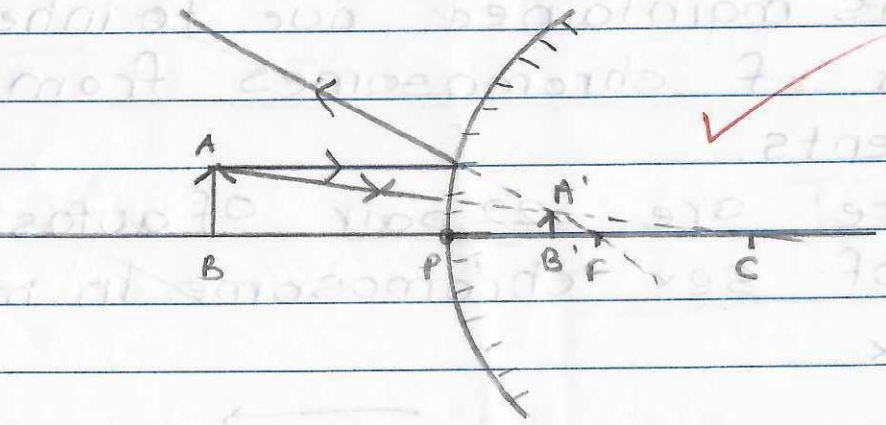
(i)



Position of object → At infinity

Nature of image → virtual, erect, diminished,
At focus on other side

(ii)



Position of object → Between infinity and Pole

Nature of image - Virtual, Erect, diminished, between F and P on other side

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Q.30)

(a) There are total 23 pairs of chromosomes present in humans or 46 chromosomes are present.
Out of them only 1 pair of chromosomes or only 2 chromosomes are sex chromosomes.
In male → X Y
In female → X X

(b)(i) In sexually reproducing organisms, the number of chromosomes is maintained due to inheritance of equal number of chromosomes from both the parents.

(ii) In humans, there are 22 pair of autosomes and 1 pair of sex chromosome. In male - X Y and female X X.

→

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(iii) The gametes ~~half~~ have only half the amount of DNA ~~of~~ and chromosomes present in them.

(iv) Hence, when ~~these too~~ the ~~two~~ male and female gamete combine during fertilisation, Original number of chromosomes is attained

(v) A diploid zygote is formed (2n) from haploid gametes (n).

Q.31)

(a) The hormone 'x' is adrenaline and the gland that secretes them is adrenal glands

(b) Adrenaline hormone prepares the body for fight or flight. There are 3 main process that take place →

i) It increases the heart rate

ii) It decreases blood supply to skin and digestive system due to contraction of

muscles around small arteries in these organs and diverts blood to skeletal muscles.

(iii) It increases breathing rate due to contraction of rib cage and diaphragm.

Q.32)

(a) Aim - To show iron articles get rusted only under specific conditions of moisture and air.

Materials required - Iron nails, three testube, calcium chlorid, water, oil

Procedure -

(a) Take three test tubes A, B and C

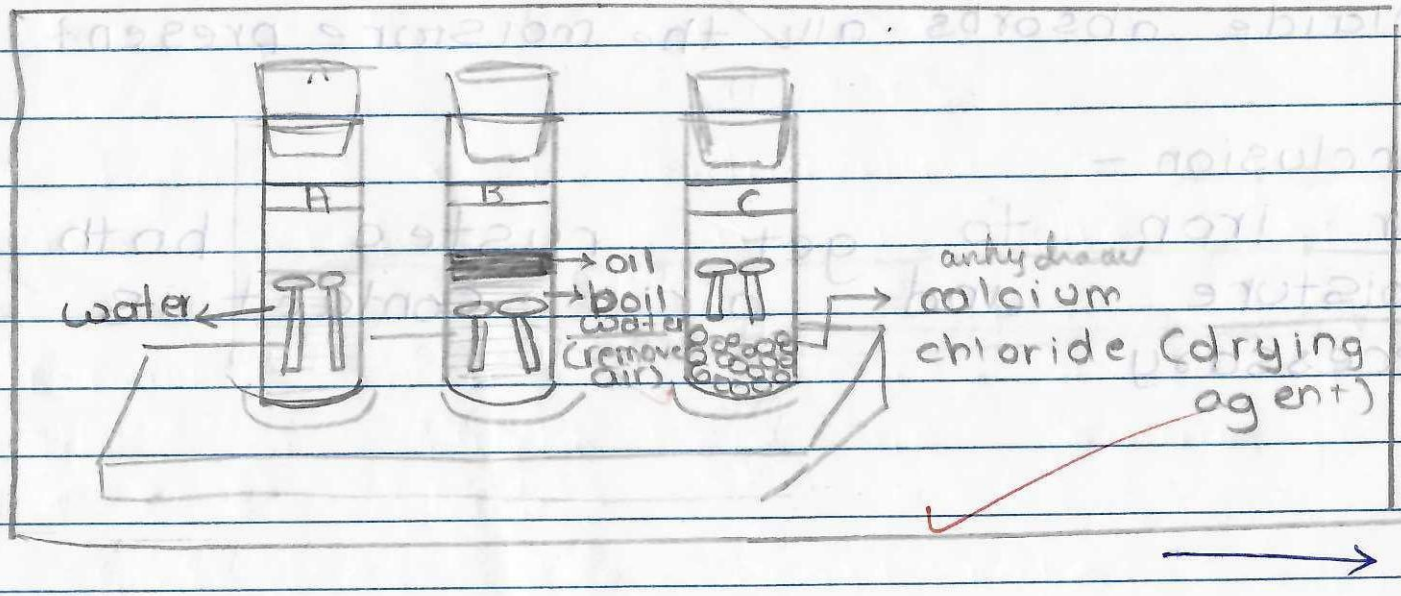
(b) In test tube A, pour some water and put iron nails in it and cork it.

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(c) In test tube B, put iron nails in boiled distilled water and on top a layer of oil. and cork it

(d) In test tube C, put calcium chloride along with iron nails and cork it

(e) Do the experimental set up asshown below →



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Observation -

i) In test tube A, iron nails develop red- brown flaky substance (rust)

ii) In test tube B and C no rust is observed as in test tube B boiled distilled water and layer of oil prevent air to interact with iron nail. In test tube C, calcium chloride absorbs all the moisture present.

Conclusion -

For iron to get rusted both moisture and air contact is necessary.

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Q.33)

(a) ~~(a)~~ Displacement reactions in which a high reactive metal displaces less reactive metal from its salt soln can be used to obtain two metals from middle of reactivity series → Manganese and Iron

(I) Reacⁿ of Aluminium with manganese oxide-
Here Al being more reactive displaces Mn.

$4Al + 3MnO_2 \longrightarrow 2Al_2O_3 + 3Mn + \text{heat}$

The heat evolved is so large that Mn is obtained in molten form

(II) Reacⁿ of ~~Fe₂~~ ferric oxide with alumin-
-ium to obtain Iron →
Here Al replaces iron from its oxide.

$Fe_2O_3 + Al \longrightarrow Fe + Al_2O_3 + \text{heat}$

→ This is also known as thermit reacⁿ. ⟶

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(b) Metals high up in the reactivity series cannot be obtained by reduction of ~~metal~~ oxides by carbon as -

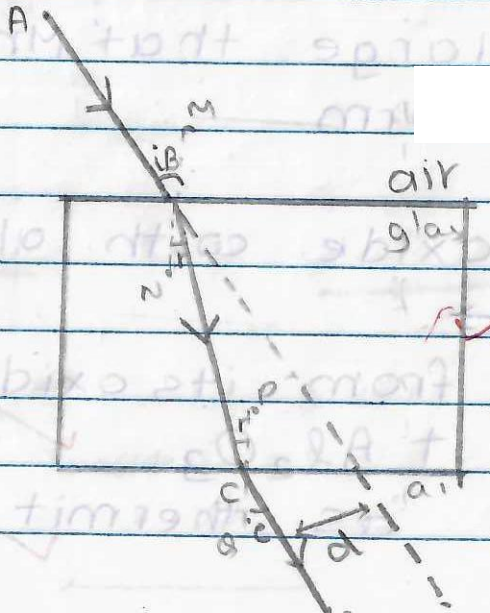
(i) These metals (K, Na, Ca) are highly reactive

(ii) They have more affinity for oxygen rather than carbon.

Section - D

Q.34)

(b)(i)



Here,

AB - ~~re~~ incident ray

$\angle ABM$ → angle of incidence

$\angle NBC$ → angle of refraction

BC → refracted ray

$\angle QCP$ → angle of emergence

CP → emergent ray

d → lateral displacement

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(II) Snell's law of refraction of light → The ratio of sine of angle of incidence to the sine of angle of refraction is constant for a given colour and given pair of media. ($i < 90°$)

$$\frac{\sin i}{\sin r} = \text{constant}$$

This constant is known as refractive index of medium 2 with respect to medium 1.

(III)

Aspect	Virtual image by convex lens	Virtual image by concave lens
(I) Object distance	Between focus and lens of mirror / optical centre	Anywhere from infinity to optical centre
(II) Magnification	Greater than 1 Image is enlarged	Less than 1 Image is diminished

Q.35)

(i) (I) Ovary →
Produces female gametes, eggs which take part in the the process of fertilisation.

(II) Fallopian tube →
It is the site of fertilisation of female gamete (egg) and male gamete (sperm).

(III) Uterus →
Provides nutrients for nourishment of foetus. It has thick and spongy lining which provides necessary conditions if egg is fertilised. During pregnancy, Placenta develops which helps to provide nutrients from mother's body and removal of waste from body of foetus.

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(ii) Two contraceptive methods used by males →

(a) Mechanical barriers – condoms

→ These help to prevent sperm transfer to female body and prevents transmission of STD's (Sexually transmitted ~~devices~~ ~~des~~ diseases)

~~(b) Intra-uter~~

(b) Surgical method → vasectomy

→ In this the vas deferens of male is blocked, which prevents sperm transfer.

Q.36

(b)

(i) Carbon forms compounds mainly by covalent bonding because —

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Carbon is tetravalent (2,4)
It either has to gain or lose e^-.
Hence it forms covalent bonding by sharing of e^-

(a) If ~~carbon~~ carbon loses $4e^-$ to form C^{4+} cation, then it will be difficult for nucleus to hold on to $2e^-$ with $6p^+$. A lot of energy will be lost

(b) If carbon gains $4e^-$ to fom C^{4-} anion, then the nucleus will have $10e^-$ with just $6p^+$ which is unstable. This process also requires a lot of energy.
Hence carbon forms compounds by covalent bonding as there are also other properties which allow it to do so like catenation, tetravalency and small size.

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(II) Covalent compounds have low melting and boiling point as they have weak intermolecular forces of attraction which can easily be overcome due to heat

(III)

(I) Covalent compounds don't have any free electrons, they only constitute of shared pair of electrons. Hence they are bad conductors of electricity.

(II) Carbon shows self-linkage of property of forming long chain of carbon atoms. It forms saturated, unsatured compounds. It can even form branched, cyclic chains. Due to its tetravellency, it forms bonds with other carbon atoms.

Section-E

Q.37)

(a)

Wire	Insulation cover
Live wire	Red
Neutral wire	Black
Earth wire	Green

(b) Given →

$P = 1000\ \omega$

$V = 220\ V$

To find → I

$\Rightarrow P = VI$

$\Rightarrow I = \frac{P}{V}$

$\Rightarrow I = \frac{1000}{220} = 4.5$

$I = 4.5A$

Hence current rating should be atleast 5A.

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(c)

(i) Earth coire is connected to the metallic body of appliances to prevent overfloco of current in a circuit and to prevent fatal accidents and damaging of appliance. The earth coire is connected to a metal plate buried inside earth. Whenever there is an excess of current flowing through circuit, it offers loco resistance path for current to floco in earth therefore saving appliances such as electric iron from damage and prevent overloading /breaking of circuit.

Q.38)

(a) $2NaCl + 2H_2O \xrightarrow{\text{electrolysis}} 2NaOH + H_2 + Cl_2$

(Caustic Soda) (Hydrogen) (Chlorine)

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(b)	Products obtained	uses
i)	H_2 (hydrogen)	fuels, ammonia for fertilisers, margarine
ii)	NaOH (caustic soda)	De-greasing metal, soap and detergent, artificial paper making
iii)	Cl_2 (Chlorine)	PVCs, disinfectant, swimming Pool

(c)(ii) Water of crystallisation -
Fixed no. of water molecules present in one formula unit of salt.
e.g. (i) Washing soda → $Na_2CO_3.10H_2O$
Sodium carbonate decahydrate

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(ii) $CuSO_4 \cdot 5H_2O$
(copper sulphate)

(iii) $FeSO_4 \cdot 7H_2O$
(ferrous sulphate)

Q.39)

(a) → Photosynthesis is the process by which green plants prepare their own food.

→ Photosynthesis is the the process by which some green plants and bacteria prepare their own food by using inorganic substances like carbondioxide and water in presence of sunlight and chlorophyll to form glucose.

(b) $6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$
glucose.

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(c)

(i) In desert plants ⟶

(a) Take up Carbon dioxide at night

(b) Prepare an intermediate product

(c) This product is acted upon by energy absorbed by chlorophyll during the day time.

(d) After this they split water to form hydrogen and oxygen and Reduction of carbondioxide to carbohydrates.

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